AuthentiMark: Production Deployment & Ethical AI Guide

Comprehensive Architectural Blueprint, Hosting Strategy & Security Hardening Matrix

1. System Architecture Overview

AuthentiMark is a deep-learning system for imperceptible neural image watermarking and AI provenance detection. It integrates a **FastAPI + PyTorch backend** (executing Autoencoder, Variational Autoencoder, and Vision Transformer models) and a high-performance **React + Vite Single-Page Application (SPA)**. Production deployment follows a **Decoupled Topology**: serving the static SPA via global Edge CDNs while deploying containerized PyTorch workers on dedicated ML compute nodes.

2. Hosting Platform Selection & Comparison

Target Layer	Recommended Platform	Key Technical Advantages	Cost / Constraints
Frontend (React SPA)	Vercel / Netlify / Cloudflare	Global Edge CDN caching, automated Git CI/CD, instant SSL, zero cold-starts.	Free tier is production-ready.
Backend (ML Engine)	Hugging Face Spaces (Docker)	Native PyTorch ecosystem, direct support for multi-GB `.pt` checkpoints, persistent storage.	Free CPU; persistent hardware from $0.05/hr.
Backend (API Instance)	Render / Railway	Automated Docker builds, automatic restart on OOM/crash, secret environment variables.	$5–$7/mo for 24/7 dedicated server.
Enterprise Cloud	AWS ECS / GCP Cloud Run	Auto-scaling container clusters, custom VPC security, dedicated GPU instances (e.g. AWS g4dn).	Pay-per-second; requires DevOps setup.

3. Production Dockerfile & Checkpoint Distribution

Model Storage Strategy: Pretrained weights (ae_latest.pt, vae_latest.pt, detector_latest.pt) are excluded from Git via .gitignore. They should be hosted on AWS S3 or Hugging Face Hub, downloading at cold-start. **Optimized PyTorch Build:** Installing CPU-only PyTorch wheels reduces image size from >5 GB to <850 MB.

```
FROM python:3.10-slim
WORKDIR /app
RUN apt-get update && apt-get install -y --no-install-recommends build-essential && rm -rf /var/lib/apt/lists/*
RUN pip install --no-cache-dir torch torchvision --index-url https://download.pytorch.org/whl/cpu
COPY backend/requirements.txt requirements.txt
RUN pip install --no-cache-dir -r requirements.txt
COPY backend/ backend/
COPY models/ models/
EXPOSE 8000
CMD ["uvicorn", "backend.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

4. Ethical AI Governance & Content Provenance Policy

Watermarking and synthetic media detection carry critical societal implications. Production deployment requires proactive safeguards against false accusations, surveillance creep, and adversarial tampering.

1. Probabilistic Transparency	Vision Transformers operate probabilistically. Never present detection verdicts as absolute proof. The UI must present calibrated metrics (e.g. *'Likely Watermarked (98.4%)'* or *'Inconclusive'*) alongside explicit human review advisories.
2. Zero-Data Retention (Privacy)	Strict ephemeral memory policy: all uploaded images must be processed exclusively in RAM streams (io.BytesIO) and discarded immediately after inference. No user image data is saved to disk or used for retraining.
3. Anti-Stripping Safeguards	Attack simulation benches can inadvertently assist bad actors in identifying exact noise/compression thresholds required to strip watermarks. Implement rate-limiting on attack queries and rotate secret message seeds.
4. Demographic & Domain Invariance	Ensure watermark perturbation imperceptibility and detection sensitivity remain uniform across diverse human skin tones, medical scans, line art, and varying camera color profiles.

5. Security Architecture & Threat Mitigation Matrix

Threat Category	Attack Vector / Exposure	Production Mitigation Control
Denial of Service (DoS)	Computational exhaustion from concurrent PyTorch inference runs.	Enforce IP-based rate limiting via slowapi (max 30 req/min). Constrain worker threads and enforce strict request timeouts.
Decompression Bomb	Malicious gigapixel images causing server Out-Of-Memory (OOM) crashes.	Enforce Image.MAX_IMAGE_PIXELS = 20_000_000 and enforce a 10 MB file size upload limit at the reverse proxy layer.
Model Inversion & Probing	Adversary sending repeated queries to compute gradients and forge watermarks.	Add subtle non-linear quantization to returned confidence scores. Detect and throttle high-frequency automated probe scripts.
CORS & Transport Security	Unauthorized cross-origin API abuse or man-in-the-middle data snooping.	Restrict FastAPI CORSMiddleware strictly to the production frontend domain. Enforce TLS 1.3 encryption across all endpoints.

6. Pre-Flight Production Launch Checklist

[ ] Environment Binding: VITE_API_URL environment variable mapped to live backend API domain.
[ ] CORS Whitelist: Wildcard allow_origins=['*'] removed; restricted solely to production frontend URL.
[ ] Model Integrity: Checkpoint hashes verified upon download from secure cloud bucket.
[ ] Transport Encryption: Full TLS 1.3 / HTTPS certificates active on both frontend and backend.
[ ] Ephemeral Verification: Audited storage layer to guarantee zero user image retention on server disk.
[ ] Rate Limiter Enabled: slowapi active and stress-tested against brute-force attacks.